Example 12.24

Urn-I contains 8 red and 4 blue balls and urn-II contains 5 red and 10 blue balls. One urn is chosen at random and two balls are drawn from it. Find the probability that both balls are red.

Solution

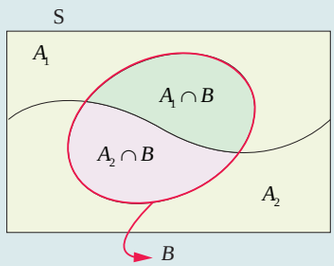
Let A_1 be the event of selecting urn-I and A_2 be the event of selecting urn-II.

Let B be the event of selecting 2 red balls.

We have to find the total probability of event B. That is, $P(B)$.

Clearly A_1 and A_2 are mutually exclusive and exhaustive events.

	Red balls	Blue balls	Total
Urn-I	8	4	12
Urn-II	5	10	15
Total	13	14	27



We have

$$P(A_1) = \frac{1}{2}, \quad P(B/A_1) = \frac{8c_2}{12c_2} = \frac{14}{33}$$

$$P(A_2) = \frac{1}{2}, \quad P(B/A_2) = \frac{5c_2}{15c_2} = \frac{2}{21}$$

We know

$$P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$$

$$P(B) = \frac{1}{2}.\frac{14}{33} + \frac{1}{2}.\frac{2}{21} = \frac{20}{77}.$$

Example 12.25

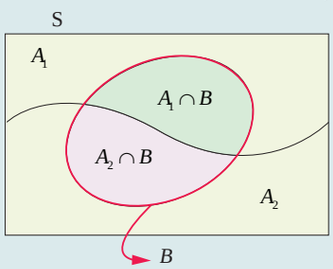
A factory has two machines I and II. Machine-I produces 40% of items of the output and Machine-II produces 60% of the items. Further 4% of items produced by Machine-I are defective and 5% produced by Machine-II are defective. If an item is drawn at random, find the probability that it is a defective item.

Solution

Let A_1 be the event that the items are produced by Machine-I, A_2 be the event that items are produced by Machine-II. Let B be the event of drawing a defective item.

We have to find the total probability of event B. That is, $P(B)$.

Clearly A_1 and A_2 are mutually exclusive and exhaustive events.



Therefore, $P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$

We have $P(A_1) = 0.40, \quad P(B/A_1) = 0.04$

$P(A_2) = 0.60, \quad P(B/A_2) = 0.05$

$$P(B) = P(A_1) \cdot P(B/A_1) + P(A_2) \cdot P(B/A_2)$$

$$= (0.40)(0.04) + (0.60)(0.05)$$

$$= 0.046.$$

12_11th-Maths-Vol2-Unit12.indd 260 16-12-2022 11:11:47